

Singularities

Key Points

A matrix which is not invertible is often called singular as its transform creates a singularity collapsing dimensions

Basic Skills

Q1. Find the following determinants

a) $\begin{vmatrix} 8 & -4 \\ 3 & 6 \end{vmatrix}$ b) $\begin{vmatrix} 3 & -2 & -5 \\ 0 & 2 & 3 \\ 4 & 3 & 2 \end{vmatrix}$

Q2. Determine if the following matrices are singular

a) $\begin{pmatrix} 4 & 1 \\ 8 & 2 \end{pmatrix}$ b) $\begin{pmatrix} 2 & -15 \\ -1 & 5 \end{pmatrix}$ c) $\begin{pmatrix} 4 & 4 & -2 \\ 5 & -6 & 1 \\ 3 & -2 & 0 \end{pmatrix}$

Q3. For a 2D transform $ad = bc$ means the transform is singular, explain how you can see this identity when looking at the matrix using the transform $\begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix}$ as an example

Competency Skills

Q4. For the singular 2D transform $M = \begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix}$

- a) Find the point which the line $y = x$ is mapped to
- b) Find the line which all 2D space is mapped (collapsed) to
- c) Geometrically explain why an inverse transform cannot exist
- d) Find the line which maps to the point (5,-10)

Q5. Given that the matrix $\begin{pmatrix} 4 & k & -2k \\ 5 & -6 & 1 \\ 3 & k+1 & 0 \end{pmatrix}$ is singular find k

Q6. For the transform $M = \begin{pmatrix} 2 & 3 \\ 6 & 9 \end{pmatrix}$ determine the line of singularity and find any invariance (points or lines) that exists under the transform

Advanced Skills

Q7. Which singular matrix transform collapses 2D Euclidean space to 0D Euclidean space

Q8. For the matrix $M = \begin{pmatrix} 3 & -1 & 2 \\ -1 & 0 & 2 \\ 4 & -2 & 8 \end{pmatrix}$

- a) Show that it maps all points to a single plane and find that plane
- b) Explain why the matrix equation $M \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ will never have a unique solution for all constants a, b, c
- c) Furthermore give a condition on constants such that $M \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ will have infinite solutions and find the line mapping when $a, c = 1, b = 2$

Extension

Q9. A particle is circling around the unit circle when the space is transformed by a singularity collapsing 2D space to the origin over some time, in what motion would you expect the particle to travel to the origin